Midmarket Context: ‘Magic Quadrant for Hyperconverged Infrastructure’

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Initiatives: Midsize Enterprise IT Leadership

Hyperconverged infrastructure is evolving into the de facto standard for on-premises and hybrid cloud integration for many midsize enterprises. MSE I&O leaders should leverage this report to identify the HCI vendors and products best suited to their scale and requirements.

Get the full story

This content is part of a larger body of research on this topic.

Magic Quadrant for Hyperconverged Infrastructure

Market Differentiators

Finite staffing levels (typically 10 to 30) and budgets (typically less than $20 million) necessitate that midsize enterprise (MSE) infrastructure and operations (I&O) leaders champion investments that are rightsized, cost-effective and emphasize operational simplicity. In a majority of cases, hyperconverged infrastructure (HCI) will yield better total cost of ownership (TCO) than traditional three-tier approaches within midsize data center environments, due primarily to the operational efficiency resulting in operating expenditure (opex) savings and capital expenditure (capex) avoidance. Native capabilities such as centralized management, data protection, disaster recovery (DR), replication, deduplication and compression within HCI solutions often meet or exceed MSEs’ functional requirements.

A December 2018 poll at Gartner’s Infrastructure, Operations and Cloud Strategies Conference in Las Vegas found that of the MSE infrastructure and operations (I&O) leaders surveyed:

- 46% have HCI deployed in production/test/development environment.
- 5% were using HCI exclusively in production.
- 52% have IT organizations of fewer the 50 full-time equivalents (FTEs).

The “versatilist” nature of MSEs’ IT teams (see “How to Build Digital Dexterity in Infrastructure and Operations”) dictates that a premium be put on service and support quality, reliability, and ease of administration.

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While some attributes are consistent across the spectrum of MSEs, such as high levels of virtualization (80% to 90%) and standardization on a single hypervisor and/or storage array, the infrastructure footprints within these organizations remain diverse (see Table 1).

Table 1: MSE Server and Storage Footprints

	25%	Median	75%	Average
Server Count	35	76	140	106
Storage (TB)	15	40	100	87
n = 177				

Source: Gartner Midsize Enterprise Summit End User Survey 2018

MSEs require the ability to run mixed application workloads with guaranteed performance per cluster, which has resulted in MSEs opting for all-flash HCI deployments. With Nonvolatile Memory Express (NVMe) storage going mainstream, along with increased drive capacity, HCI has the capacity and performance capabilities to effectively run all the virtual workloads within a typical MSE environment.

Although the "system hardware requirement" has been dropped from the Magic Quadrant criteria, MSE I&O leaders continue to disproportionally favor single-vendor HCI appliance products with robust native capabilities. This results in simpler, less expensive vendor management and maintenance, and relinquishing the need to integrate disparate products.

Considerations for Technology and Service Selection

The hyperconverged infrastructure market is experiencing 45% annual growth, offering MSE I&O leaders mature products from major server OEMs/original design manufacturers (ODMs) and software companies. While entry-level pricing options differ, MSE I&O leaders must judge candidate HCI solutions in the context of solution requirements and operational preferences.

MSE I&O leaders should seek an HCI vendor that can deliver:

- Simplified pricing, maintenance, contracts and licensing, with an all-inclusive pricing scheme.

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- High-quality and responsive pre- and postsales support. Ideally, the price should not drastically affect the quality of support, including SLAs for rapid hardware replacement.
- Preferred hypervisor support and native integration to cloud providers — that is, hybrid cloud integration, and management capabilities that align with your organization's hybrid cloud vision.
- A focus on the MSE sector — that is, a vendor that delivers a full-featured product within a scaled-down footprint with a unified and intuitive management interface.
- Integration with preferred/incumbent:
  - Infrastructure management tools (such as VMware vCenter and Microsoft System Center Configuration Manager [SCCM]).
  - Legacy storage products and archival data platforms, which can ease the transition to software-defined solutions.
- Robust native, backup, data protection and DR capabilities, including orchestration.
  - Appliance-to-appliance, DR as a service (DRaaS) service, or partnerships with proven data protection vendors.
  - Overall reduced risk solution, including more-secure hybrid environments.

MSE I&O leaders who prefer specific hardware platforms or who wish to maintain relationships with resellers offering HCI software preinstalled on hardware platforms within their portfolio should use Gartner's HCI Magic Quadrant and Critical Capabilities to guide decision making.

MSE I&O leaders should *not assume* the best choice is one that is in the Leaders quadrant, but should give preference to cultural fit and vendors specializing in midsize enterprise deployments, specific geographies or verticals, and *should not* discount vendors in the Niche Players or Challengers quadrants.

MSE I&O leaders should:

- Check the number of reference customers and reviews whose implementations are of similar scale. Seek to speak to at least three references with success stories and ask the provider for a reference that did not work to understand potential missteps.

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- Conduct an RFP with a least three vendors (see "Toolkit: Sample RFP for Hyperconverged Infrastructure").
- Learn from MSE early adopters that have experienced performance issues typically relating to network issues. In hindsight, most consider a network evaluation a prerequisite for a successful HCI implementation.
- Focus on TCO benefits, such as savings from virtualization licensing, made achievable by hypervisor license portability, and labor savings that result from ease of management.
- Clearly identify the expected benefits such as operational efficiency (opex) and cost (capex).
- Realize that vendor lock-in is inevitable at MSE scale.

Before MSE I&O leaders sign a contract, they should fully understand the pricing, licensing and maintenance costs involved to help ensure a lower TCO and delivery of expected benefits (see "How to Derive the Total Cost of Ownership of Hyperconverged Integrated Products").

Notable Vendors

Vendors included in this Magic Quadrant Perspective have customers that are successfully using their products and services. Selections are based on analyst opinion and references that validate IT provider claims; however, this is not an exhaustive list or analysis of vendors in this market. Use this perspective as a resource for evaluations, but explore the market further to gauge the ability of each vendor to address your unique business problems and technical concerns. Consider this research as part of your due diligence and in conjunction with discussions with Gartner analysts and other resources.

Cisco

HyperFlex combines Cisco's HX Data Platform (software) with its Unified Computing System (UCS) server platform, with integrated fabric networking. Sold as a fully integrated appliance, HyperFlex supports VMware vSphere and Microsoft Hyper-V hypervisor, and allows existing UCS customers to integrate their installed base into HyperFlex deployments.

HyperFlex is available as a bundle with Cisco CloudCenter, a cloud management platform. HyperFlex with CloudCenter can be used to manage on-premises vSphere deployments, and work with a wide range of Cisco data center solutions, as well as public cloud options such as Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP). CloudCenter and Cisco Intersight (cloud-based system management) are designed to enable hybrid IT strategies and deliver operational simplicity when deploying, optimizing and managing workload in multicloud environments and/or on-premises.

Key MSE Differentiators:

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- MSE who prefer one-stop shopping and support across various IT domains can utilize HyperFlex as a fundamental building block in Cisco's Data Center Anywhere portfolio for seamless integration of multicloud orchestration, software-defined WAN (SD-WAN), core networking, and so on.
- Cisco has a rich partner ecosystem, including validated solutions with Cohesity, Commvault and Veeam to address secondary storage, DR and data protection needs.
- Strong pre and postsales support and a vast reseller community make HyperFlex compelling to MSEs that wish to maintain existing reseller relationships.
- Cloud-based management (Intersight) of hybrid infrastructure aligns to MSE operational environments.

Dell EMC

VxRail is the Dell EMC flagship HCI offering, leveraging VMware's HCI vSAN software product. VxRail is available as a fully integrated appliance series for a variety of use cases (such as ROBO, edge and data center). Dell EMC and VMware jointly develop VxRail appliances, with co-engineering effort on the VxRail management software.

VxRail Manager comes preinstalled with a dashboard for automating VxRail deployment and configuration, and monitoring the health of the entire cluster. Level 1 and Level 2 support are provided by Dell EMC, with Level 3 assistance provided by VMware, although all interactions are still delivered to the customer through Dell EMC.

Key MSE Differentiators:

- Dell EMC lowered the entry point and price point to allow MSE clients to start with a two-node VxRail cluster.
- Dell EMC has broadened PowerEdge server model and media choices for HCI appliance and simplified the administrative experience and life cycle management through VxRail Manager Automation enhancements.
- Dell EMC has significantly improved its pre- and postsales support for VxRail throughout 2018 by leveraging its robust direct sales as well as its channel partners worldwide.
- VxRail is the VMware integrated hyperconverged appliance with VMware Cloud Foundation support, enabling it to be future-proof for next-generation VMware hybrid cloud technologies.

Hewlett Packard Enterprise (HPE)

Hewlett Packard Enterprise's (HPE's) SimpliVity hyperconverged appliances are based on mature, feature-rich and scalable HCI software, and deployed at thousands of enterprises of all sizes. In addition to medium and large all-flash models, the HPE SimpliVity offers midmarket solutions in

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single- and dual-socket variations; flexible configurations resulting in multiple cost-effective options for midsize enterprises, remote, branch and edge deployments.

The solution supports VMware and Hyper-V hypervisors and is designed to consolidate multiple workloads on one platform. HPE offers SimpliVity HyperGuarantee for a set of features that enable efficiency, availability and recovery services for its HCI customers. HPE provides MSEs with the alternative of pay-per-use via HPE GreenLake Flex Capacity, enabling customers to adopt an opex model for their HCI services.

Key MSE Differentiators:

- HPE SimpliVity has high storage efficiency and robust data reduction services with always-on global deduplication and compression leveraging the HPE OmniStack Accelerator Card or in a software-only version.
- Guaranteed enterprise-class resilience features are offered, such as hardware high-availability, instant VM recovery, built-in backup solution, and HPE SimpliVity RapidDR automating site-to-site disaster recovery.
- Easy-to-navigate choices of all flash robust HPE appliances are available in two models: HPE SimpliVity 380 and HPE SimpliVity 2600 for environments requiring dense form factors.
- HPE has been focused on delivering and supporting its HCI solutions to midsize enterprises globally through its broad reach of channel partners.

Nutanix

Nutanix continues to lead the HCI market with its innovative Enterprise Cloud Platform, which leverages Nutanix's native AHV Virtualization or third-party hypervisors with Nutanix's software-defined storage, compute, data security, software-defined networking and management software. Customers can acquire Nutanix as software for deployment on customer-procured servers or as an appliance with preinstalled Nutanix software through resellers and server suppliers (Lenovo, Dell EMC, Fujitsu and IBM).

Over the past year, Nutanix introduced new software capabilities and hardware models targeting edge and ROBO deployments. The vendor also released new products, including software-based data encryption at rest, Nutanix Calm for application orchestration, Flow for software-defined networking and security, Files for unstructured data, Xi Beam for multicloud governance and cost control, Xi Frame as a desktop-as-a-service offering and Xi Epoch for multicloud application monitoring.

Key MSE Differentiators:

- Nutanix delivers a robust management and self-service management interface in Prism, and supports hybrid cloud through Xi Leap for disaster recovery as a service in the Americas (additional geographies will become available throughout 2019).

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- Midsize enterprises will find Nutanix AHV Virtualization (based on KVM) to be an attractive lower-cost alternative hypervisor to VMware ESXi, with a comparable feature set and easy migration path.
- Nutanix has high customer satisfaction, starting from presales to support experiences.
- Nutanix provides midsize enterprises with ongoing consolidation opportunities by aggressively adding products, features and capabilities to its portfolio of services.

Pivot3

Pivot3's HCI Acuity Platform and its policy-based management features (performance quality of service [QoS], data protection, data security and QoS scheduling) and nonvolatile memory express (NVMe) flash caching and tiering enable the running of multiple, mixed application workloads. In 2018, Pivot3 introduced Acuity X3 Edge/Micro Datacenter, an entry-level small form factor offering with fully redundant three-node configurations starting at approximately $55,000, with support pricing ranging from 6% to 18%, based on support level (Standard, Premium or Business-Critical). Single-node pricing starts at $18,000.

Pivot3 intelligent hybrid cloud and Cloud Edition for Amazon Web Services provide multiple options to leverage AWS for storage replication and backup. MSE I&O leaders can connect Pivot3's HCI to public cloud providers (Google, Amazon and Microsoft [Azure]) via an instance of Pivot3 software running in the cloud and/or supported third-party cloud connector providers. Intelligent hybrid cloud also includes automated data and workload mobility between on-premises and the cloud.

Key MSE Differentiators:

- Pivot3's automated policy-based management software enables the performance and security consistency MSEs desire to confidently run multiple and mixed application workloads on a single HCI solution.
- Acuity X3 Edge/Micro-Datacenter (typically deployed at the edge and centrally managed through vCenter) with Pivot3's standard Acuity X5 product at the core provides MSEs the flexibility to appropriately size their environments to meet their requirements.
- Pivot3 Cloud Edition for Amazon Web Services provides simplifies cloud integration for customers, making it easier for MSEs to leverage and consume cloud services.
- Pivot3's approach to data protection can significantly reduce the raw storage capacity requirements of an HCI solution, lowering acquisition costs and improving TCO for midsize enterprises.
- MSE customers have praised Pivot3's pre- and postsales support, and ease of use.

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Scale Computing

Scale Computing's HCI product (HC3) and vision are rooted in delivering full-featured, affordable solutions coupled with administrative simplicity. HC3, delivered as an appliance with a Kernel-based Virtual Machine (KVM)-based hypervisor, is offered with cost-effective premium-only support (12% to 14% of the capital purchase price), and typically represents the lowest acquisition cost of the more mature HCI vendors. HC3's self-healing machine intelligence automatically identifies, mitigates and corrects problems in the infrastructure in real time, which eliminates the need for infrastructure management componentry.

Scale Computing's hybrid cloud integration connects on-premises and cloud infrastructure and data via its HC3 Cloud Unity solution. Scale Computing's "ScaleCare services" deploys a remote HC3 instance on Google Cloud Platform and creates a gateway to connect to on-premises infrastructure via a virtual extensible LAN (VXLAN). To secure hybrid cloud workloads, Scale Computing has partnered with WinMagic (SecureDoc CloudVM) and HyTrust (DataControl) to ensure MSEs can move applications freely and securely in the cloud or on-premises.

Key MSE Differentiators:

- Scale Computing prioritizes and actively pursues midsize enterprises, though not exclusively; hence, the majority of Scale Computing's deployments and customers are midsize enterprises.
- The HC3 product family has a highly transparent, all-inclusive pricing structure, with MSE customers giving its pre- and postsales support a high rating.
- Scale Computing is focused on pursuing and maintaining partnerships that support traditional backup solutions. The HC3 Cloud Unity DRaaS is an easy, affordable and secure way to replicate and run an HCI cluster on Google Cloud Platform
- Scale Computing offers fully functional single-node solutions, capable or replicating to a central site or cloud, starting at $8,000; and HC3 edge with multicluster management to address remote and branch office requirements.

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Summary Translation: Innovation Insight for the Cloud Center of Excellence

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